

Rutwik Kadam

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PROFILE SUMMARY

Full Stack AI & Backend Engineer specialized in Generative AI & High-Performance Systems. Proven track record in reducing infrastructure costs by 90% through Edge AI (WebAssembly/ONNX) and optimizing data latency by 40%. Deeply proficient in Python, ML Algorithms, and FastAPI, with hands-on experience in LLM orchestration (DeepSeek, Cohere), Vector Databases (Qdrant), and Real-time Voice AI.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL
AI/ML & Data Tools: PyTorch, Pandas, NumPy, LangChain, LangGraph, LangSmith, RAG Optimization (Hybrid Retrievers, ColBERT).
Generative AI: Prompt Engineering, LoRA, PEFT, Quantization, Transformer (Encoder-Decoder), DeepSeek, Mistral.
Backend & APIs: FastAPI, FastRTC, REST APIs, WebSockets.
Database: Vector Databases (Qdrant, ChromaDB), MongoDB, SQLite.
DevOps & Cloud: AWS, Docker, GitHub Actions, CI/CD, MLOps, Hugging Face Inference Endpoints.
Frontend: React.js, React Native, HTML5, CSS3, Tailwind CSS.

WORK EXPERIENCE

AI Engineer | NG LABS (Nagpurukul) August 2025 – Present

Project 1: Speech-to-Speech System [link](#)

- Trained a production-grade, end-to-end **Speech → Text → Speech** model in English, Hindi, Gujarati, Indian English accent and deployed across multiple products including an AI Interviewer and Voice Assistant.
- Trained on 50+ hours of audio; achieved <10ms speech synthesis latency with a Word Error Rate (WER) of 10%.
- Enabled **voice cloning** with fewer than 20 audio samples, making it highly data-efficient.
- Quantized model supports deployment on low-end / edge devices.
- Reduced speech synthesis cost from **\$18/hr → \$0.12/hr** (~99% cost reduction).
- Tech Stack:** Python, PyTorch, ONNX, Typescript for inferencing.
- Deployment and deployment:** HuggingFace, Sagemaker, Azure.

Project 2: Zoe AI Interviewer [link](#)

- Built an AI-powered mock interview platform that conducts interviews, evaluates responses, and provides personalized post-interview feedback.
- Designed a RAG (Retrieval-Augmented Generation) pipeline using Qdrant to retrieve context from a user's past interview history, enabling the AI to identify weak areas and ask targeted follow-up questions.
- Optimized the system flow to under 1 second, supporting **10,000 concurrent interviews** with an end-to-end STS **latency of <3 seconds** — outperforming **OpenAI, Google Gemini, and ElevenLabs**.
- Orchestrated multi-step AI agent workflows using LangGraph for robust state, memory, and context management across interview sessions.
- Tech Stack:** Python, FastAPI, LangGraph, Pandas, React
- Deployment and Database:** AWS EC2, Amplify, Qdrant, Firebase.

Software Developer | InternetSoft Jan 2025 – July 2025

Project 3: Autonomous QA — Multi-Agent Testing System

- Engineered a **multi-agent QA automation system** that autonomously tested a **POS (Point of Sale)** platform, detected bugs, and reported them directly to **Zoho** — eliminating manual reporting.
- Designed a **3-stage agentic orchestration pipeline**:
- Assigner Agent** — analyzed bug severity & developer expertise to route issues optimally
- Reviewer Agent** — triggered on GitHub commits to validate fixes before merge
- Production Agent** — performed final cross-validation of fix, review report & code diff before deployment
- Reduced **manual QA intervention by 60%**, accelerating the end-to-end development cycle.
- Tech Stack:** Python, LangChain, LangGraph, RAG, Foundation Model

Education

- Masters in Computer Science | MIT.**
Aug 2023 – July 2025 | CGPA: 8.73/10
- Bachelors in Computer Science | Modern College of Science.**
July 2020 – May 2023 | CGPA: 9.22/10